

SUMMARY

Ph.D. candidate in Economics at the University of Texas at Austin with an M.S. in Statistics. Specialize in causal inference, pricing, demand estimation, and structural industrial organization using large-scale observational data. Experience spans antitrust litigation, policy research, machine learning, and production software engineering.

EDUCATION

- University of Texas at Austin

Ph.D. Economics (expected 2026); M.A. Economics

University of Arizona

M.S. Statistics; B.A. Mathematics and Computer Science

Austin, TX

Aug 2020–Present

Tucson, AZ

Aug 2013–May 2018

SKILLS

- Programming:

Economics & Statistics:

Tools:

Languages:

Python, R, Julia, SQL, TypeScript, Rust, Java, C, Stata, MATLAB, Bash

Causal inference, structural modeling, demand estimation, machine learning (XGBoost), nonparametrics, numerical optimization

Pandas, tidyverse, Postgres/PostGIS, spatial analysis, Docker, Git, HPC, Make/Snakemake, L^AT_EX

English (native); Spanish (proficient)

PROFESSIONAL EXPERIENCE

- Texas Office of the Attorney General

Economics Fellow

World Bank Group

Short-Term Consultant

JPMorgan Chase & Co.

Software Engineer

Austin, TX

Jul 2024–Jun 2025

Remote

Feb 2021–Aug 2021

Houston, TX

Jun 2018–Jun 2020

Analyzed antitrust cases involving algorithmic collusion, online advertising auctions, and digital markets using industrial organization methods.

Drafted economic analyses and reviewed expert rebuttals supporting litigation strategy.

Identified a security vulnerability affecting a key litigation matter, informing defensive legal strategy.

Presented technical findings to attorneys and translated quantitative analyses into accessible legal arguments.

Built statistical indices measuring political participation in Afghanistan using large-scale survey data.

Developed algorithms for ordinal survey aggregation and PCA-based index construction.

Authored technical documentation and presented methods to economists and policy researchers.

Developed and led internal training on R, tidy data, functional programming, and reproducible workflows.

Developed and maintained production trading systems in Python and TypeScript supporting interest rate derivatives.

Designed regulatory workflows and trade models for new financial instruments.

Migrated legacy platform to microservice architecture using React, GraphQL, and WebSockets.

Collaborated with traders, product managers, and engineers to deliver production software.
- SELECTED RESEARCH
- Dealer Consolidation and Vertical Relationships:

Dynamic Pricing in Automotives:

Demand Elasticity and Curvature (with Eugenio Miravete):

Causal analysis of automobile dealership consolidation using transaction-level data and geospatial analysis to estimate effects on prices and product availability.

Developed and estimated a dynamic structural model of manufacturer-dealer relationships. Combined structural estimation with machine learning (XGBoost) to recover underlying allocation and sales target policy of the manufacturer. Use estimates to recover welfare effects of dealer allocation policy.

Nonparametric estimation of consumer demand to study the relationship between elasticity, curvature, pass-through, and market power, with comparisons against widely used demand systems.
- ADDITIONAL EXPERIENCE
- Teaching:

Research Computing:

Teaching assistant for Master’s Econometrics, Industrial Organization, and Causal Inference. Led weekly review sessions, designed programming assignments, and taught statistical computing.

Led workshops on high-performance computing for economics Ph.D. students and developed departmental guidance for parallel computing in Julia, Python, and MATLAB.